

Pipeline Steps

- **Read the YAML metadata.** From each YAML file we take the burst name, DM, central frequency c_{freq} , total bandwidth bw, the reference burst time t_{ref} , and any zap-channel information. The zap-channel list is converted into frequency ranges (in MHz) that will be masked as RFI.
- **Load the voltage data.** Using the YAML paths we locate the base folder and load the two voltage streams $X(t)$ and $Y(t)$ from the matching DM-tagged .npy files.
- **Dynamic spectrum from voltages (FFT step).** The recorded data consist of two voltage time series, $X(t)$ and $Y(t)$, sampled at 336 MHz. To convert these into a dynamic spectrum, we divide the voltages into short, overlapping time segments and compute a Fourier transform for each segment. This converts the signal from the time domain into frequency channels.

For each time segment we then form the total intensity (Stokes I) as

$$I(t_k, f_j) = |X_k(f_j)|^2 + |Y_k(f_j)|^2,$$

where k labels time bins and j labels frequency channels. Consecutive spectra overlap by 50% in time, and several neighboring spectra are averaged to reduce noise.

The frequency resolution is set by the FFT length,

$$\Delta\nu = \frac{f_s}{N_{\text{FFT}}},$$

so larger FFT sizes give finer frequency resolution but coarser time resolution. In this configuration we generate spectra at 100 kHz resolution.

- **Pick ON and OFF time windows.** The ON window is the short region around the burst:

$$t_{\text{ref}} - 4 \text{ ms} \leq t \leq t_{\text{ref}} + 4 \text{ ms},$$

and the OFF window is a nearby region before the burst:

$$t_{\text{ref}} - 30 \text{ ms} \leq t < t_{\text{ref}} - 22 \text{ ms}.$$

- **Mask RFI channels using the YAML zap list.** Any frequency channels listed by the YAML zap ranges are set to NaN in both ON and OFF, and those channels are excluded from all averages and fits.
- **Remove the bandpass using OFF statistics.** For each frequency channel we measure the OFF mean and scatter:

$$\mu_{\text{off}}(f_j) = \langle I_{\text{off}}(t, f_j) \rangle_t, \quad \sigma_{\text{off}}(f_j) = \text{std}_t[I_{\text{off}}(t, f_j)],$$

then we normalize ON and OFF channel-by-channel:

$$I'_{\text{on/off}}(t, f_j) = \frac{I_{\text{on/off}}(t, f_j) - \mu_{\text{off}}(f_j)}{\sigma_{\text{off}}(f_j)}.$$

This removes the average bandpass shape.

- **Build the mean ON spectrum.** We average the normalized ON dynamic spectrum over time (using only unmasked channels):

$$I(f_j) = \langle I'_{\text{on}}(t, f_j) \rangle_t.$$

- **Compute the fractional ACF and fit for ν_s .** We form a fractional spectrum

$$x(f_j) = \frac{I(f_j) - \langle I \rangle}{\langle I \rangle},$$

and compute its autocorrelation versus frequency lag $\Delta\nu$:

$$\text{ACF}(\Delta\nu) = \langle x(f) x(f + \Delta\nu) \rangle_f.$$

The zero-lag value of the ACF contains an additional contribution from measurement noise. We estimate this noise from the OFF window and subtract its contribution from the central lag before fitting.

Lorentzian Fit to the ACF. The scintillation bandwidth is estimated by fitting a Lorentzian model to the fractional autocorrelation function of the ON spectrum:

$$\text{ACF}(\Delta\nu) = \frac{A}{1 + (\Delta\nu/\nu_s)^2} + C,$$

where ν_s is the scintillation bandwidth, A is the amplitude, and C is an offset. To reduce the influence of noisy large-lag values, the fit is performed using tapered weights that emphasize small frequency lags:

$$w(\Delta\nu) = \max\left(1 - \frac{|\Delta\nu|}{W}, 0\right),$$

where W is a scale proportional to the expected bandwidth. The parameters are obtained by minimizing the weighted residuals using a robust loss function, which further limits the impact of outliers. Central lags around $\Delta\nu = 0$ are excluded from the fit to avoid bias from residual noise correlations.

- **Power-spectrum cross-check.** We form a mean-subtracted spectrum

$$\tilde{I}(f_j) = I(f_j) - \langle I \rangle,$$

and compute its power spectrum in modulation frequency η :

$$P(\eta) = \left| \mathcal{F}\{\tilde{I}(f)\} \right|^2.$$

We then fit

$$P(\eta) \approx A e^{-2\pi\nu_s|\eta|} + C,$$

which provides an independent consistency check on ν_s .

- **Noise-corrected modulation index.** Using a Gaussian fit to the central lags of the ON autocorrelation function we estimate

$$\text{ACF}(0) \equiv G(0).$$

Let N_{on} be the number of ON time bins used in the average spectrum and let

$$\hat{S} = \langle I \rangle$$

be the mean intensity of the averaged ON spectrum (not the mean of fractional fluctuations). We apply the requested bias correction

$$m^2 \approx \max\left(G(0) - \frac{1}{\sqrt{N_{\text{on}}}\hat{S}}, 0\right), \quad m = \sqrt{m^2}.$$